

Simulation-Based Inference

Double Responses in Decision Tasks

Group 4

Ajeyaa Sinha Roy (266796)

Leanne Rosemary Monteiro (269998)

Danish Ahmad (264784)

TU Dortmund University

Department of Statistics

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1 Introduction

Speeded decision experiments, in which participants make rapid judgments in response to simple stimuli, are central in cognitive psychology. Analyses typically focus on choice and response time, with the well-known trade-off between speed and accuracy motivating the development of evidence accumulation models (EAMs).

A recurring but often overlooked behavior in these tasks is double responding: participants occasionally issue a second, opposite keypress shortly after the first. These events are more common under speed pressure and may capture post-decision dynamics and uncertainty. Evans et al. [2020] highlight double responding as an under-explored behavioral constraint for EAMs, drawing connections to partial errors, post-error adjustments, and neural evidence of ongoing accumulation. Incorporating such responses provides a novel means to reduce model mimicry, improve parameter recovery, and deepen our understanding of how decisions are finalized and sometimes revised.

In this project, we take the Racing Diffusion Model (RDM) as a baseline and extend it to account for double responses. We further consider a two-stage elevated-barrier variation, where a second press requires additional evidence beyond the initial threshold. Both models are implemented in BayesFlow, using simulation-based inference to estimate parameters on data from Evans et al. (2020).

2 Data

We used data from a lexical decision experiment by Dutilh et al. [2009], later reanalyzed by Evans et al. (2020). Four Dutch participants each completed 10,000 trials, judging whether letter strings were words or non-words. Two participants were instructed to emphasize speed, and two to emphasize accuracy. Each trial recorded the choice, response time, and whether a *double response*—a second, opposite keypress within 250 ms—occurred. For analysis, 1,000 trials per participant were randomly sampled, preserving distributions of accuracy, response times, and double-response frequencies while ensuring computational efficiency.

3 Statistical Model (Simulator)

We base our analysis on the Racing Diffusion Model (RDM), extended with explicit mechanisms for double responses. The simulator is implemented in BayesFlow via priors, a generative process, and trial-level outputs from Radev et al. [2023]

Variation 1: Standard RDM with time-window rule

Two accumulators evolve as independent Wiener processes with drifts ν_1, ν_2 . The first threshold crossing at α determines the choice and first response time $RT_1 = \min(T_1, T_2) + \tau$, where τ is non-decision time. The losing process continues; if it reaches the same threshold within cutoff $T_{\max}$, a double response is recorded with $RT_2 = \max(T_1, T_2) + \tau$, otherwise $\text{DoubleResp} = 0$.

Variation 2: Two-stage RDM with elevated barrier

Here, the first response occurs when an accumulator crosses α_1 . A double response requires the losing accumulator to reach a higher barrier $\alpha_2 = \alpha_1 + \Delta$ within $T_{\max}$:

$$RT_2 = \begin{cases} T_{\ell}^{(2)} + \tau, & T_{\ell}^{(2)} < T_{\max}, \\ 0, & \text{otherwise.} \end{cases}$$

Priors and outputs

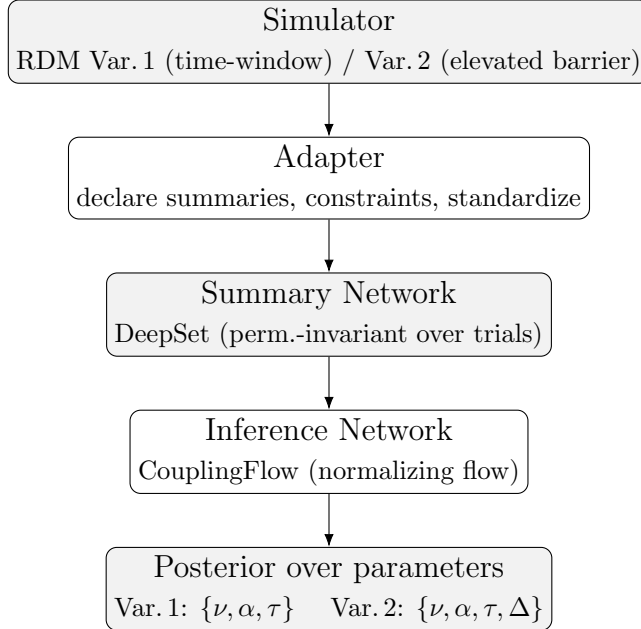
We place weakly informative priors ensuring positivity while retaining flexibility: $\nu_i \sim \Gamma(5, 0.5)$, $\alpha_1 \sim \Gamma(5, 0.2)$, $\Delta \sim \text{Exp}(s^*)$, and $\tau \sim \text{Exp}(1/0.15)$, where s^* is calibrated so that $P(\text{double response}) \approx 3\%$. Each trial produces the tuple $\{\text{Correct}, RT_1, \text{DoubleResp}, RT_2\}$. In BayesFlow, this is implemented via (i) a context function specifying the number of trials n , (ii) a prior function sampling $(\nu, \alpha, \tau, \Delta)$, and (iii) a likelihood function generating trial outcomes.

4 Approximator

We implemented amortized inference in BayesFlow using a **summary network** and an **inference network**. Trial-level simulator outputs were first processed by an **adapter**, which selected summaries, applied parameter constraints, standardized values, and formatted inputs for the networks. Summaries included correctness, first response time (RT_1), presence of a double response, and second response time (RT_2).

The two RDM variants differ only in how double responses are generated: *Variation 1* (time-window rule): a second response occurs if the losing accumulator crosses the same threshold within a cutoff. *Variation 2* (elevated-barrier rule): the losing accumulator must reach a higher threshold within the cutoff.

The **DeepSet** network mapped trial-level outputs into a permutation-invariant representation, which was then passed to a **CouplingFlow** normalizing flow. This produced flexible posterior distributions over model parameters and ensured scalability across both RDM variants.



Trial summaries: {Correct, RT_1 , DoubleResp, RT_2 }. Mechanism difference only affects how double responses are generated.

Figure 1: Simple amortized-inference pipeline for both RDM variants.

5 Training

Training was conducted in an *offline regime*, where a large number of synthetic datasets were first generated from the simulator and then used to fit the networks. For each RDM variant, we produced 2,500 training datasets, 1,000 validation datasets, and 1,000 test datasets. Each dataset consisted of multiple trials sampled from the simulator, ensuring coverage of the parameter space.

```
train_data      = simulator.sample(2500)
validation_data = simulator.sample(1000)
```

The networks were trained for up to 50 epochs with a batch size of 120, using validation loss to monitor convergence and prevent overfitting. By repeatedly exposing the networks to synthetic data from the prior, the inference procedure learned a global mapping from observed trial summaries to posterior distributions over parameters. Once training was complete, this amortized inference allowed us to directly estimate posteriors for empirical data from Evans et al. (2020) without further retraining.

```
history = workflow.fit_offline(
    data=train_data,
    epochs=50,
    batch_size=120,
    validation_data=validation_data
)
```

6 Diagnostics

We assessed both RDM variants with (i) convergence checks, (ii) simulation-based calibration (rank histograms and ECDF–difference bands), and (iii) posterior behavior (contraction and parameter recovery).

6.1 Convergence

Both variants trained stably without divergence or overfitting. Variation 1 (time-window) converged more smoothly with a slightly lower final loss; Variation 2 (elevated barrier) showed noisier validation, consistent with a higher-dimensional parameterization.

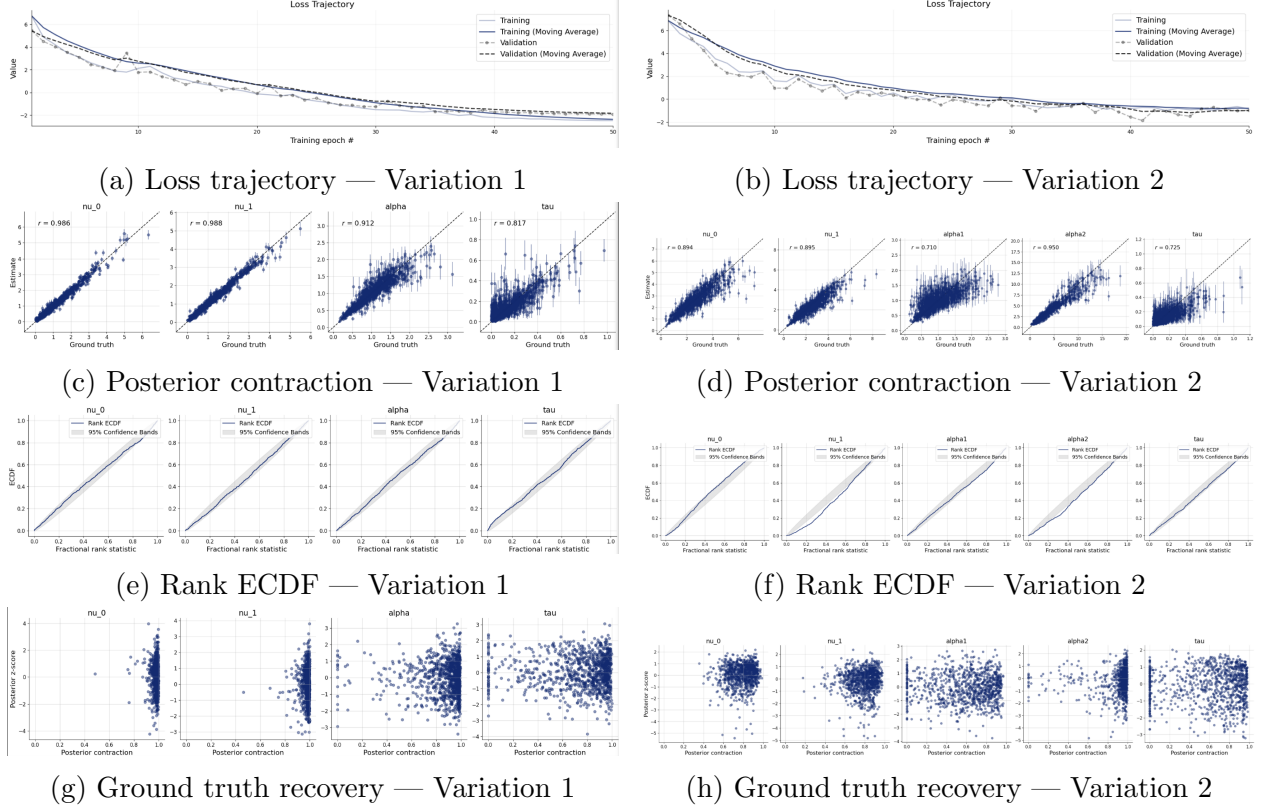


Figure 2: Training data diagnostics for both variations: (a–b) loss trajectories, (c–d) posterior contraction, (e–f) rank ECDF checks, and (g–h) ground truth recovery.

6.2 Simulation-based calibration

Posterior rank histograms. Both variants produce broadly uniform ranks. Variation 1 is closer to flat for $(\nu_0, \nu_1, \alpha, \tau)$, whereas Variation 2 shows mild deviations for ν_1 and α_1 .

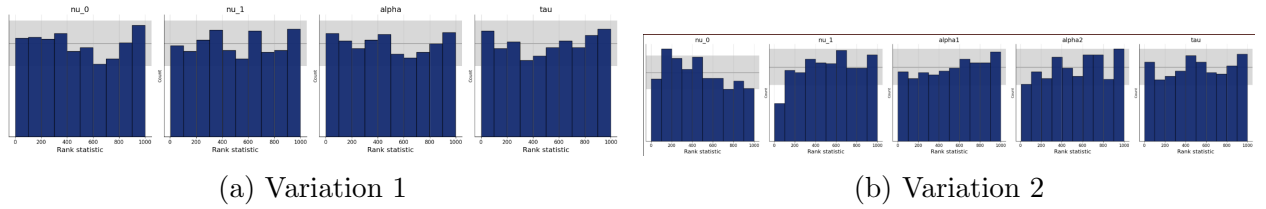


Figure 3: Posterior rank histograms (uniformity $\Rightarrow$ well-calibrated posteriors).

Posterior-predictive confidence bands. ECDF-difference curves for Variation 1 mostly lie within the 95% bands; Variation 2 exhibits more systematic departures (notably for ν_0 and α_2), suggesting mild misspecification.

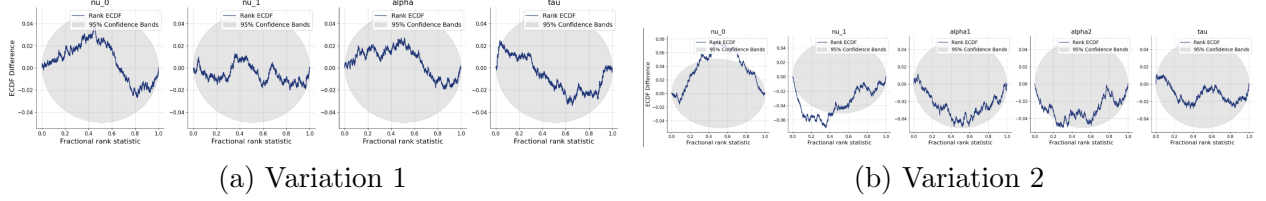


Figure 4: Posterior-predictive ECDF differences with 95% confidence bands.

6.3 Posterior Contraction and Parameter Recovery

Contraction and z-scores. Scatterplots (contraction vs. z-score) indicate strong shrinkage for ν_0, ν_1 in both variants; contraction is moderate for α (or α_2) and weakest for τ and α_1 in Variation 2, consistent with learning difficulty for early-stage/elevated barrier components.

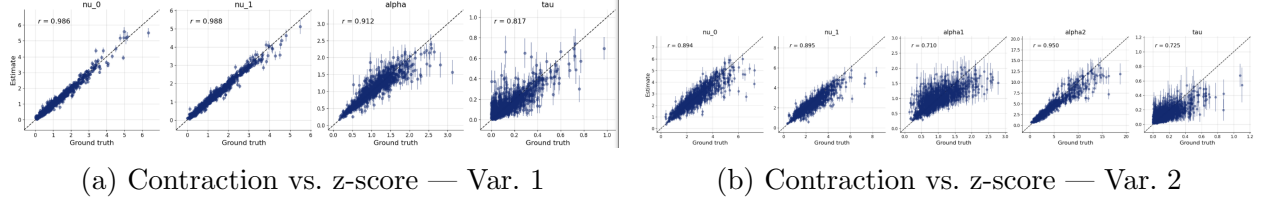


Figure 5: Posterior contraction patterns across parameters.

Recovery. Parameter recovery is excellent for (ν_0, ν_1) in both variants. Variation 1 recovers α and τ better than Variation 2, where α_1 and τ show lower correlations and wider dispersion.

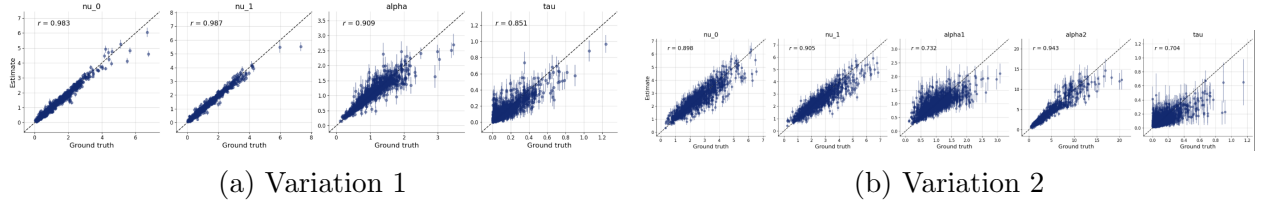


Figure 6: True vs. estimated parameters (points with posterior uncertainty).

7 Inference

Fit to Real Data

We condition the trained amortized model on each participant's observed trials and access fit with two steps. First, we inspect posterior diagnostics for the Speed (S1) parameters to verify

structure and identifiability across model variations. Then, we compare *weighted* empirical ECDFs of first-response times (correct vs. incorrect) with posterior-predictive medians and 50% intervals on a common grid.¹

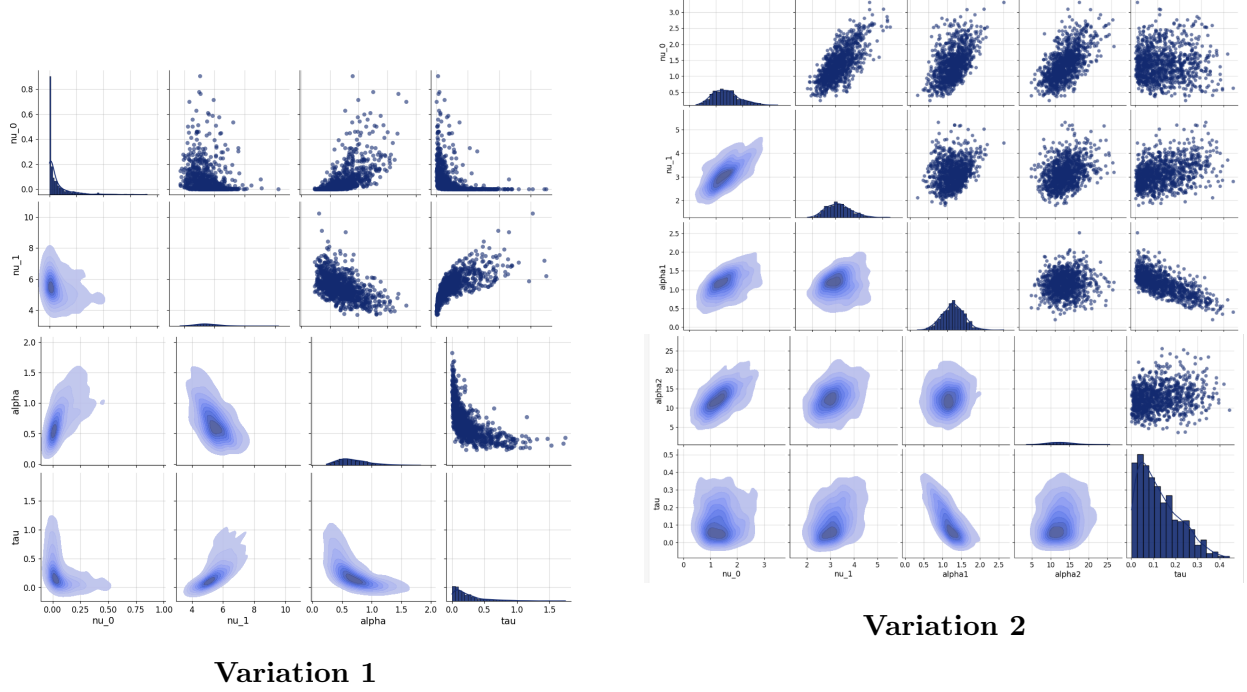


Figure 7: Posterior diagnostics for Speed (S1). Each column stacks the off-diagonal scatter matrix (top) and the corresponding contour/marginal panel (bottom).

Diagnostics for S2, A1, and A2 can be found in the Appendix A.

Variation 1: Posterior samples show stronger structural trade-offs with negative correlations, indicating tighter calibration.

Variation 2: Posteriors are more diffuse with weaker correlations, suggesting flexibility but reduced interpretability.

We now turn to posterior-predictive checks against the observed first-response times via weighted ECDFs. In addition to S1, we also perform the same ECDF analysis for Accuracy (A1), enabling direct comparison across task conditions.

¹Mapping: **list_id 3** $\equiv$ **A1** (accuracy), **list_id 2** $\equiv$ **S1** (speed). Participants within an instruction showed very similar patterns (**A1** $\approx$ **A2**, **S1** $\approx$ **S2**), so we report one representative per instruction

First-Response RTs: Accuracy (A1)

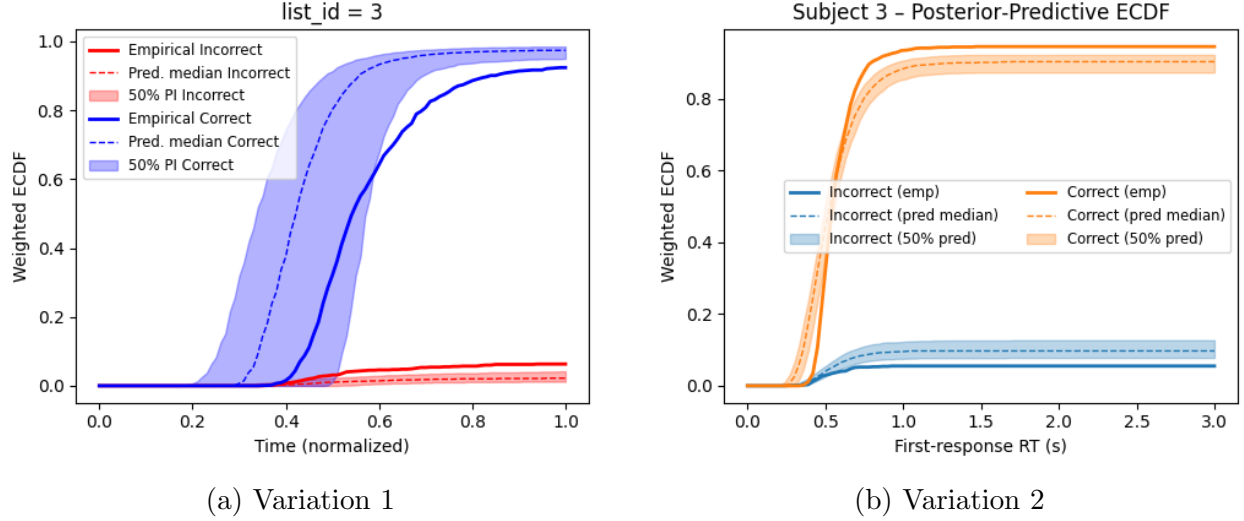


Figure 8: Posterior predictive checks for **A1**. Solid lines: empirical weighted ECDFs; dashed: posterior-predictive medians; shaded: 50% intervals.

Corresponding A2 first-response RT plots are provided in Appendix B (Figure 18).

Variation 1 The model tracks the correct-trial ECDF well, with the empirical curve largely inside the 50% band. A modest *left shift* of the predictive median (slightly faster predicted RTs) appears around mid-quantiles. Incorrect ECDF mass is small and well covered.

Variation 2 Coverage remains good and the accuracy-induced right shift in RTs is reproduced. The same mild left shift of the predictive median persists; errors are again well bracketed.

First-Response RTs: Speed (S1)

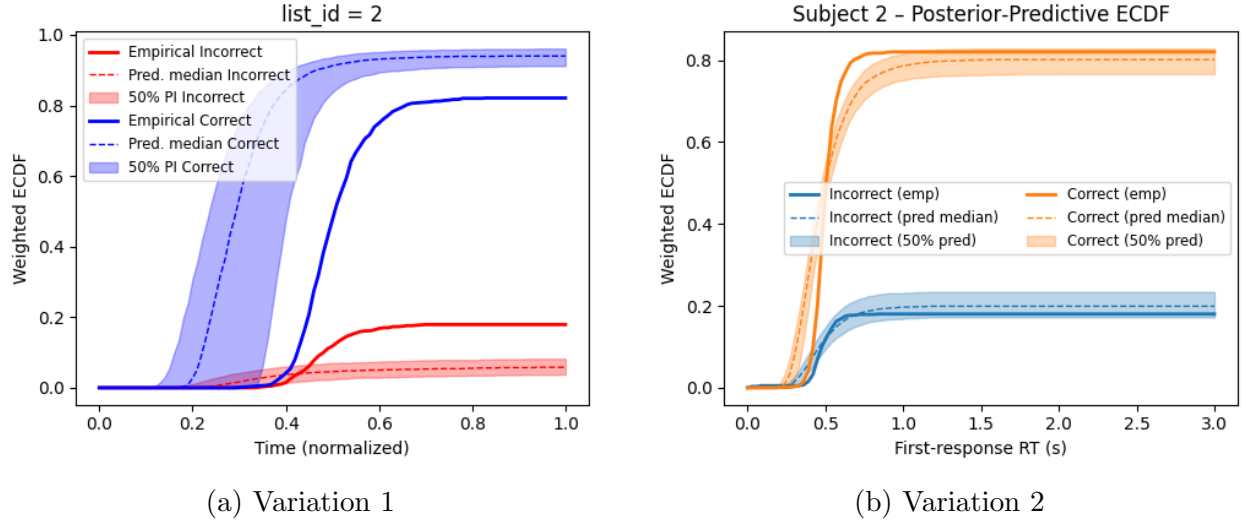


Figure 9: Posterior predictive checks for **S1**. Solid lines: empirical weighted ECDFs; dashed: posterior-predictive medians; shaded: 50% intervals.

Corresponding S2 first-response RT plots are provided in Appendix B (Figure 19).

Variation 1 The steep rise of the correct ECDF under speed instructions is captured, with empirical curves mostly inside the 50% band. A small left shift of the predictive median is visible; incorrect ECDFs are low and covered.

Variation 2 Qualitatively similar fit with a good coverage for correct and incorrect ECDFs, and a comparable mid-range left shift of the predictive median.

First-Response RTs: Accuracy vs. Speed Across both participants A1 and S1, **both variations** 1 and 2 capture slower correct RTs under accuracy, faster under speed, and the small magnitude of error ECDFs, with empirical curves largely within the 50% predictive bands. The main residual in *both A1 and S1* is a consistent, modest left shift in the predictive median for correct trials which means it predicted RTs slightly too fast around the center. **Variation 2** provides comparable coverage to Variation 1 but no clear improvement. Patterns replicate for A2 and S2.

Second-Response RTs (Double Responses): Accuracy (A1)

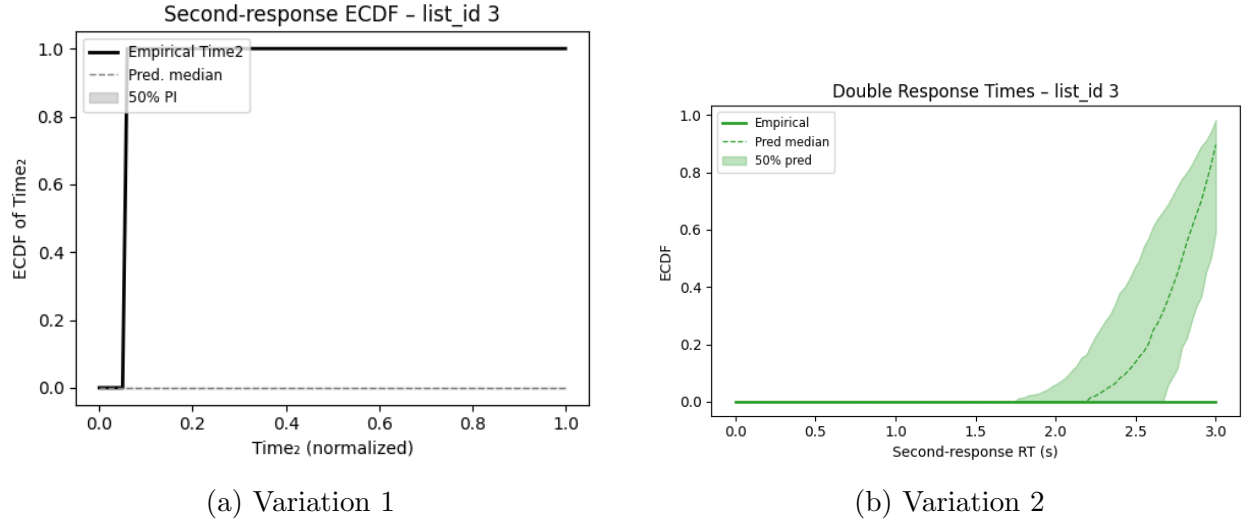


Figure 10: Posterior predictive checks for **A1** second responses. Solid: empirical ECDF of Time2; dashed: posterior-predictive median; shaded: 50% interval.

Second-response RT plots for A2 are provided in Appendix C (Figure 20).

Variation 1 Predictive mass is essentially at zero across the grid, indicating the model *rarely generates* a second press within the window. This fails to capture the empirical Time 2 distribution which shows a sharp, early rise.

Variation 2 The model now produces second responses, but concentrates them *too late* (ECDF rises mainly between $\sim 2\text{--}3\text{s}$). For A1, empirical doubles are extremely scarce and, when present, occur earlier than predicted; the model overestimates latency and frequency.

Second-Response RTs (Double Responses): Speed (S1)

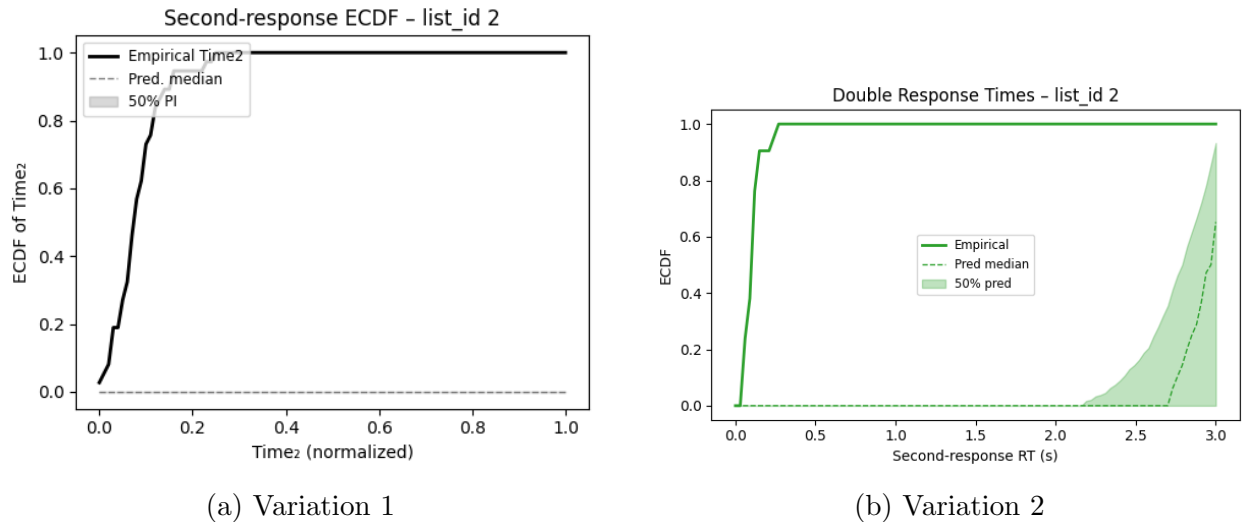


Figure 11: Posterior predictive checks for **S1** second responses. Solid: empirical ECDF of Time2; dashed: posterior-predictive median; shaded: 50% interval.

Second-response RT plots for S2 are provided in Appendix C (Figure 21).

Variation 1 As in A1, predictions are effectively zero for Time2, missing the empirical early second presses (ECDF reaches 1 well before 0.4 s).

Variation 2 Second responses are generated but are *substantially too slow*: the predictive ECDF rises only near 2.5–3 s, whereas the empirical ECDF reaches 1 by ≈ 0.3 –0.4 s. Thus, timing is strongly misaligned.

Second-Response RTs: Accuracy vs. Speed For both A1 and S1, **Variation 1** severely underfits double responses. **Variation 2** introduces second presses but places them much *later* than observed and tends to overpredict, especially under speed instructions where empirical Time 2 is concentrated very early. Hence, neither variation provides an adequate account of second-response timing.

8 Limitations and Potential Improvements

The models underfit second responses: Variation 1 failed to generate Time 2, while Variation 2 produced delayed responses. The first-response fits were fairly good, but the correct RTs were consistently too fast (left-shifted), indicating slight calibration issues, especially for τ and α_1 in Variation 2. A key limitation was the *very low base rate of double responses* in the data— $A1= 0.000$, $A2= 0.001$, $S1= 0.026$, $S2= 0.019$ —which weakens the learning signal for both frequency and timing; even with α_2 in Variation 2 the predicted double-response rate improved but remained suboptimal. Going forward, we either need more double responses with either more trials or task manipulations that raise the rate *or* stronger, domain-informed priors on second-response timing and frequency.

9 Conclusion

We fit two Racing Diffusion Model (RDM) variants to lexical-decision data with double responses using BayesFlow. Both models captured the instruction effect on first responses—faster under speed, slower under accuracy—with a small mass of errors. The empirical-vs.-predictive ECDFs displayed a systematic left shift for correct RTs. For second responses, Variation 1 essentially failed to produce Time 2, whereas Variation 2 generated second presses but predominantly too late, leaving their timing and frequency mismatched to the actual data. Overall, the models fit first-response times reasonably well, but explaining double responses will require a more detailed account of what happens immediately after the first keypress.

10 Reflection on own Learnings

We learned that BayesFlow can be utilized end to end to provide a simulator-first analysis of decision models like double-response mechanisms. In particular, building a DeepSets summary network and a CouplingFlow inference network showed us how amortized simulation based inference converts large batches of simulated trials into a reusable posterior estimator, enabling fast inference on real participants after offline training. We learned that performance depends on informative summaries (RT_1/RT_2 , Correct, DoubleResp, ECDF samples), priors that cover plausible regimes (e.g., Δ for α_2), and disciplined calibration,

posterior predictive checks and rank diagnostics to reveal issues such as the recurring left shift that implicates τ /threshold trade-offs. Using a 1,000-trial subsample per participant preserved instruction effects (accuracy and speed) and the rare second presses while keeping training tractable leaving us with a clear direction for applying BayesFlow in future projects.

A Additional Posterior Diagnostics

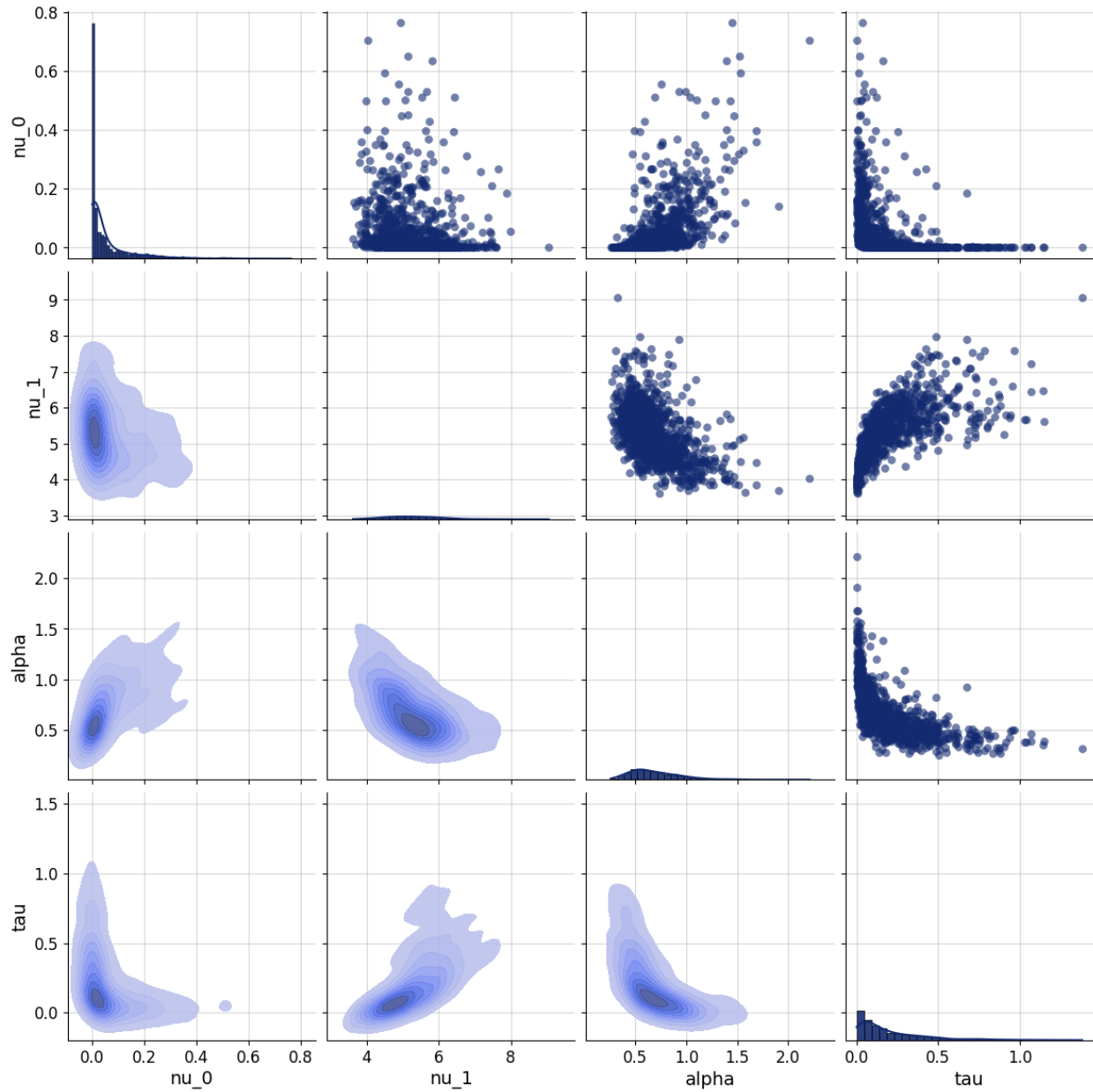


Figure 12: Posterior diagnostics for **S2**, Variation 1.

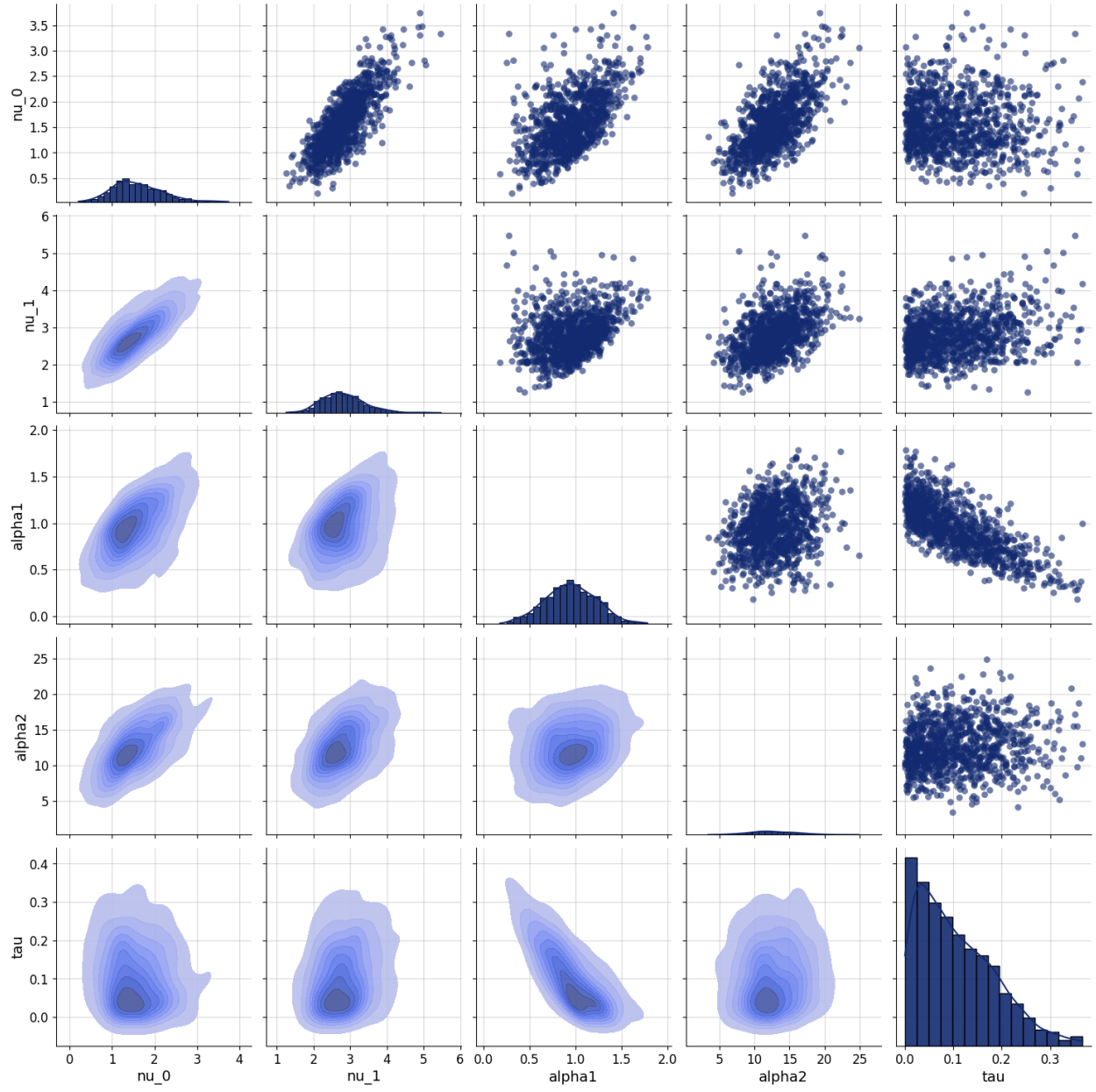


Figure 13: Posterior diagnostics for **S2**, Variation 2.

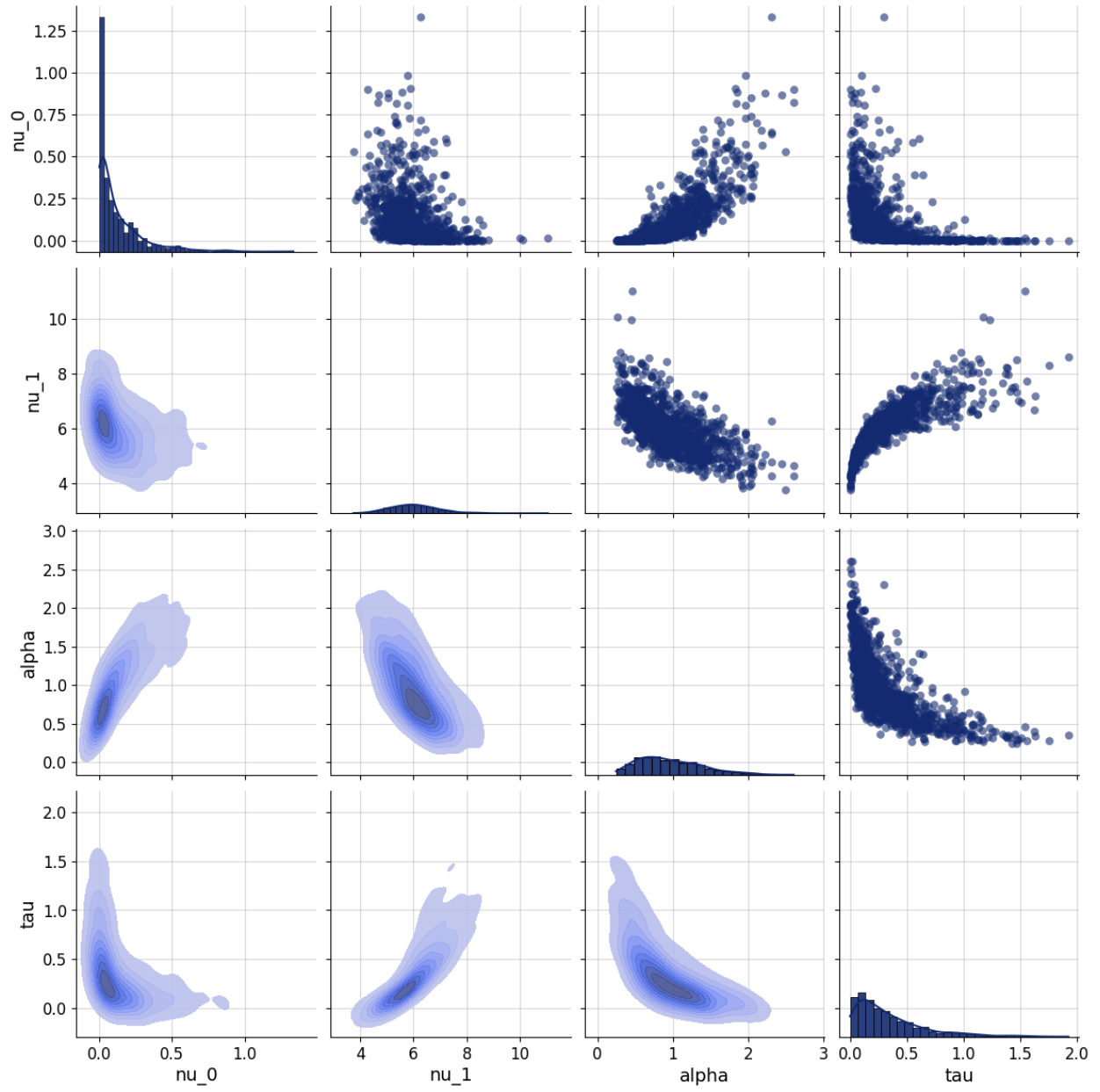


Figure 14: Posterior diagnostics for **A1**, Variation 1.

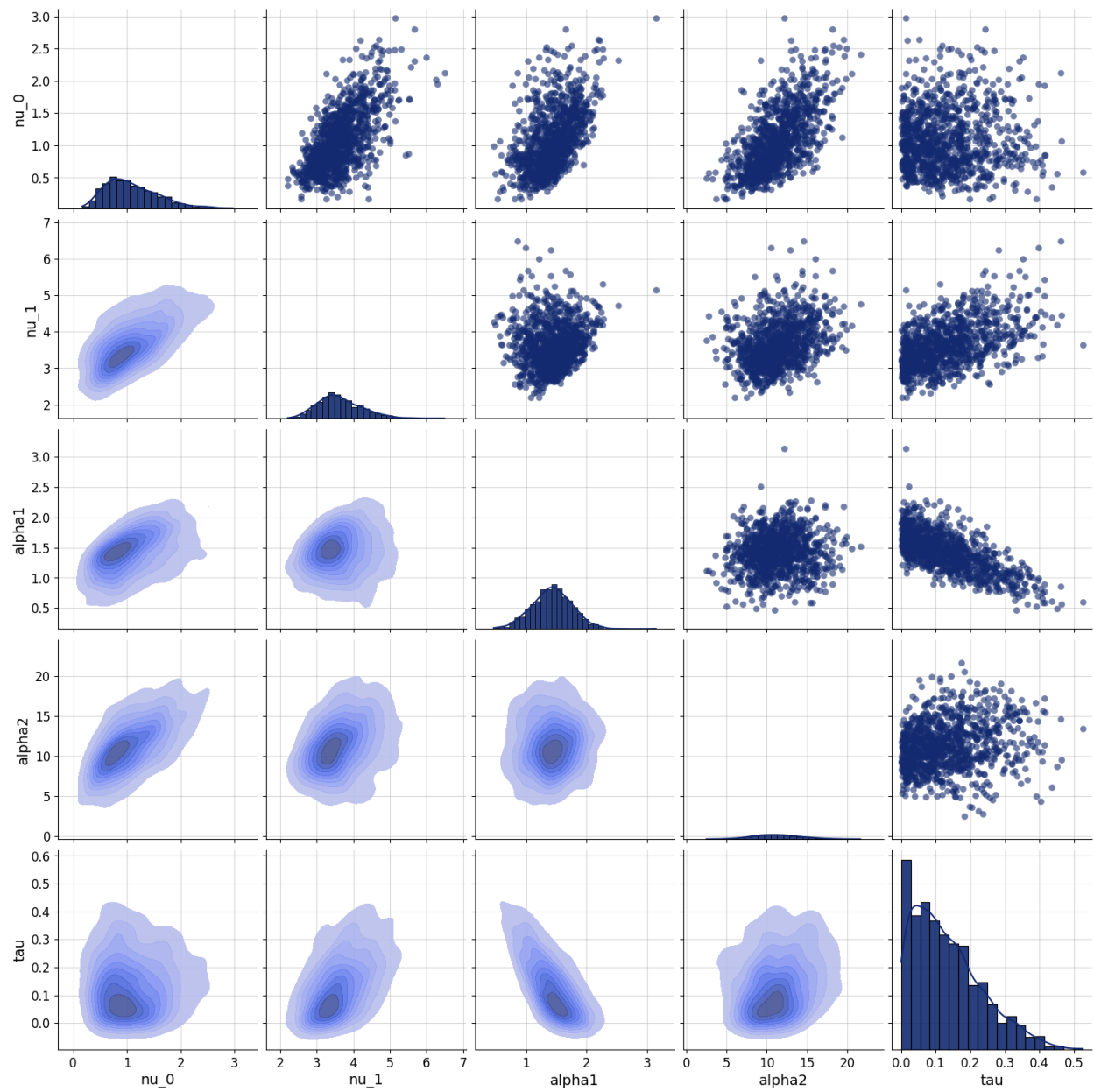


Figure 15: Posterior diagnostics for **A1**, Variation 2.

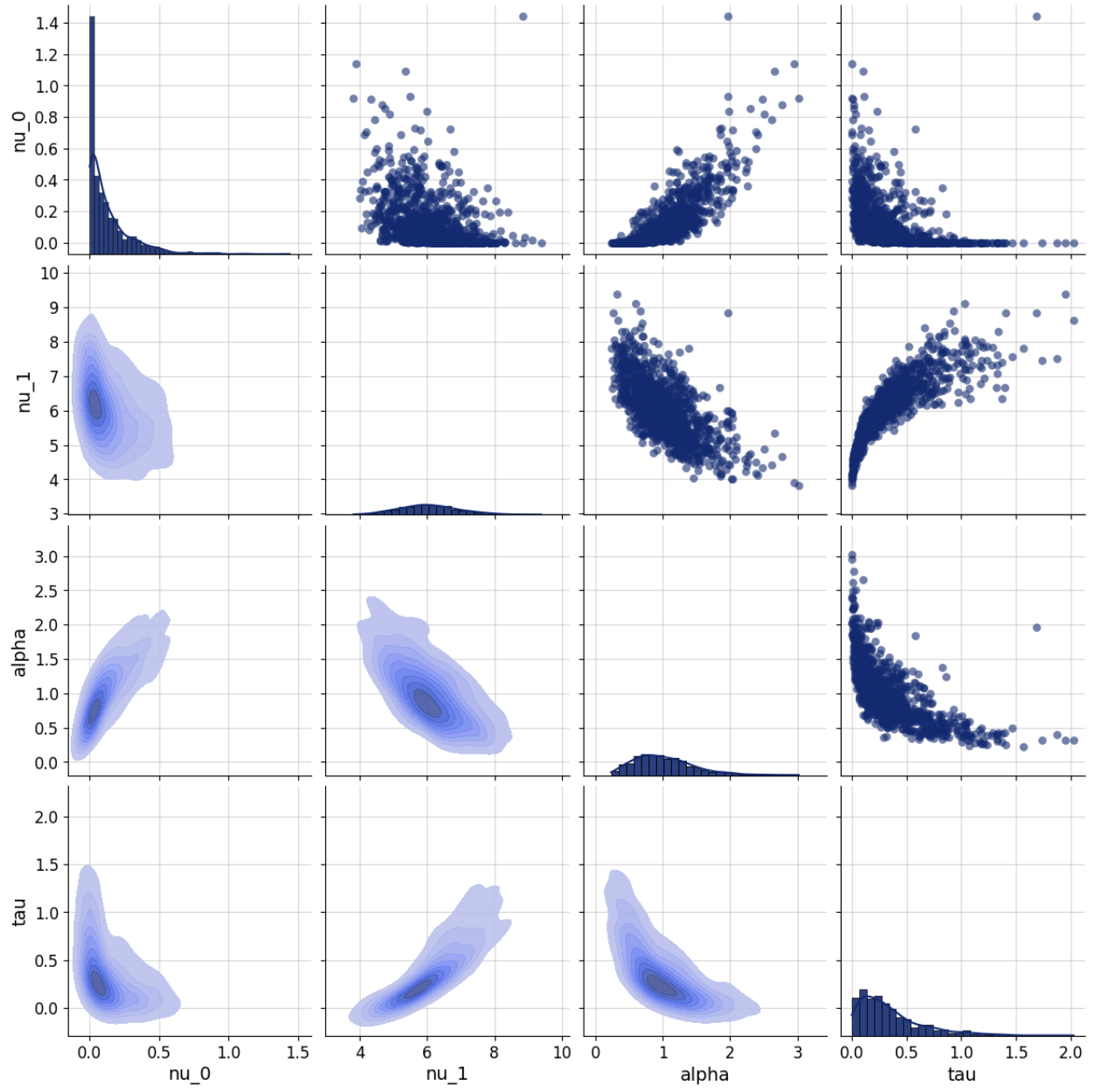


Figure 16: Posterior diagnostics for **A2**, Variation 1.

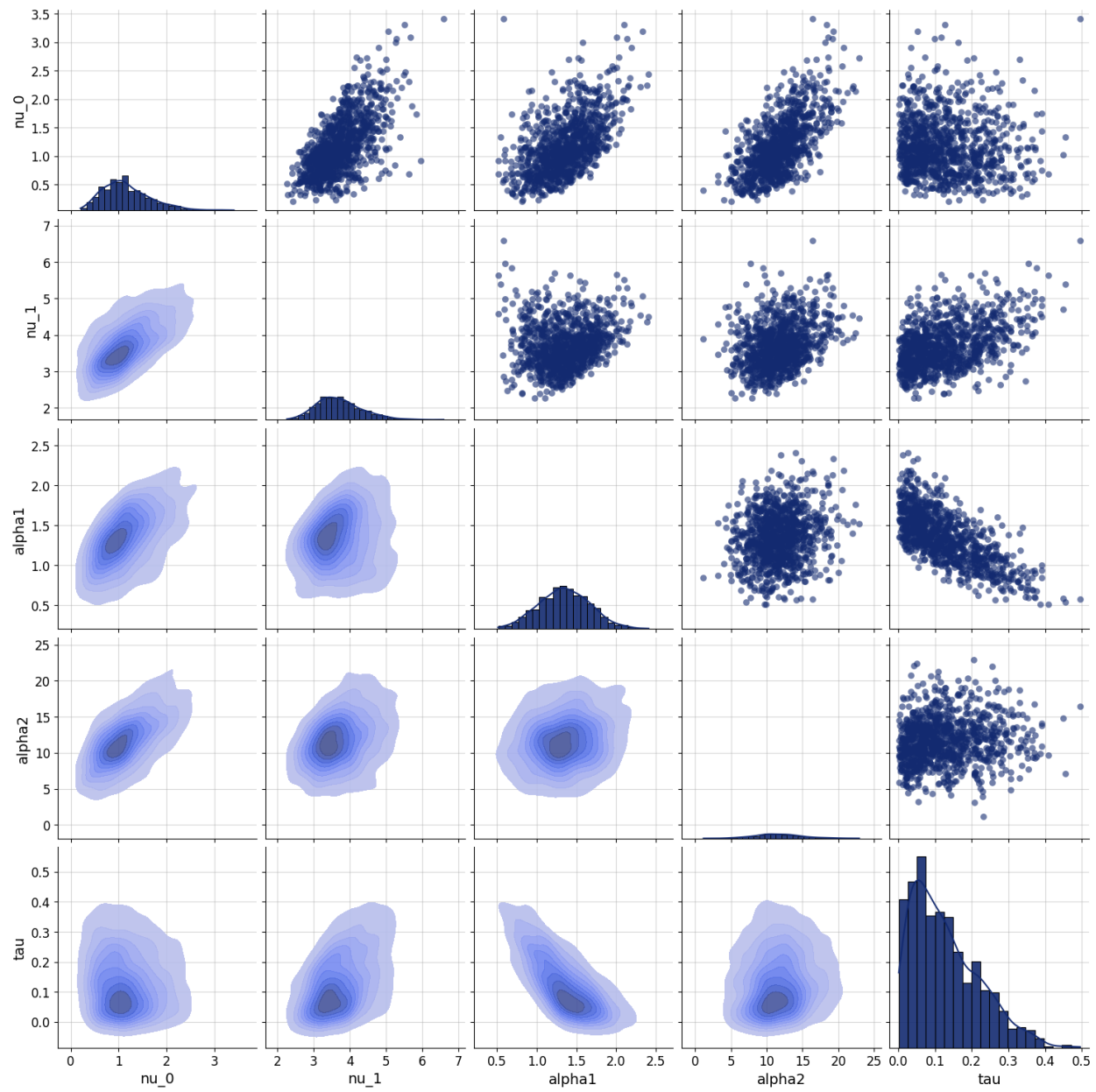
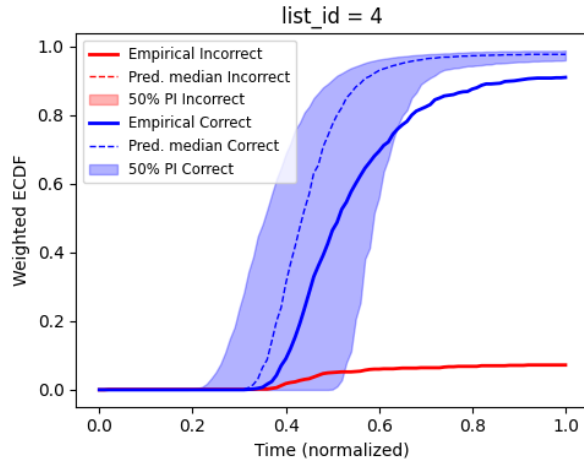
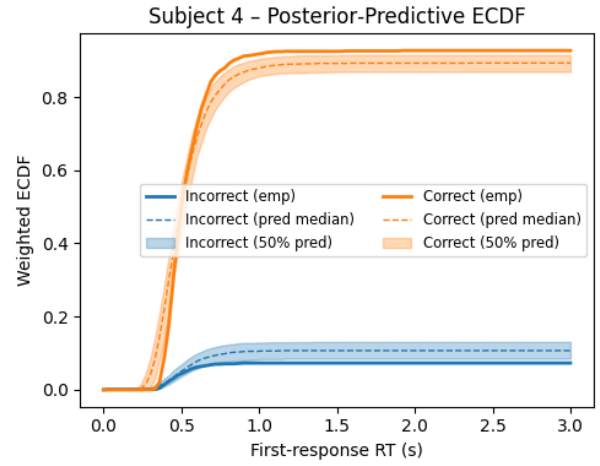


Figure 17: Posterior diagnostics for **A2**, Variation 2.

B First-Response RTs: A2 and S2

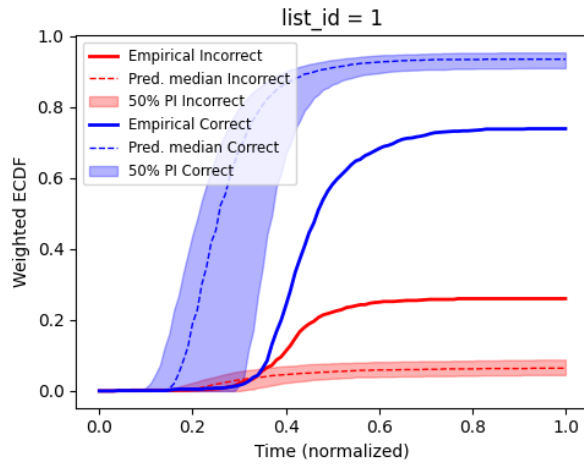


(a) Variation 1

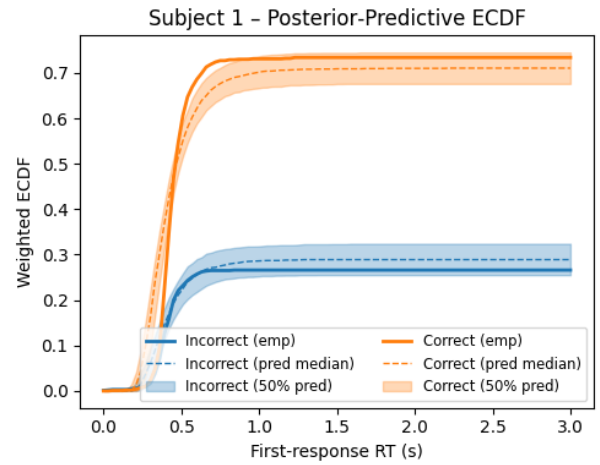


(b) Variation 2

Figure 18: Posterior predictive checks for **A2**. Solid lines: empirical weighted ECDFs; dashed: posterior-predictive medians; shaded: 50% intervals.



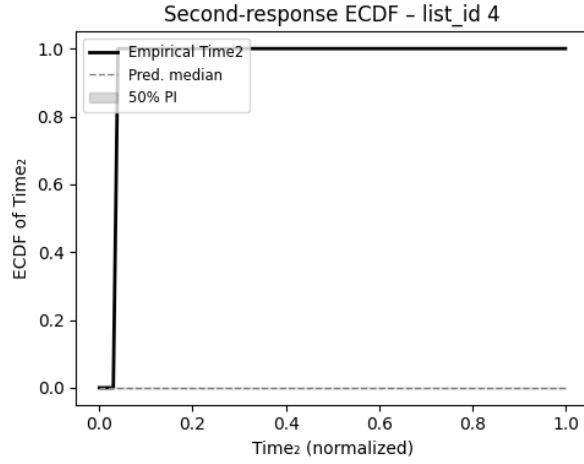
(a) Variation 1



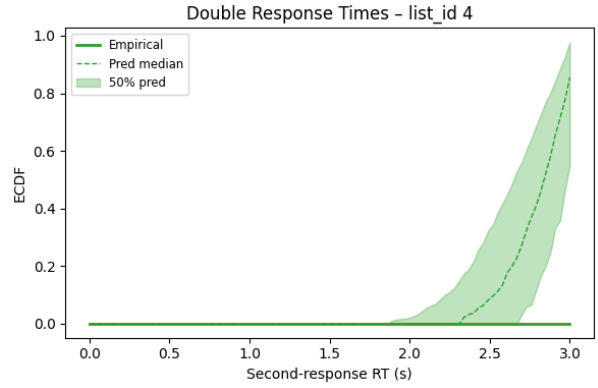
(b) Variation 2

Figure 19: Posterior predictive checks for **S2**. Solid lines: empirical weighted ECDFs; dashed: posterior-predictive medians; shaded: 50% intervals.

C Second-Response RTs: A2 and S2

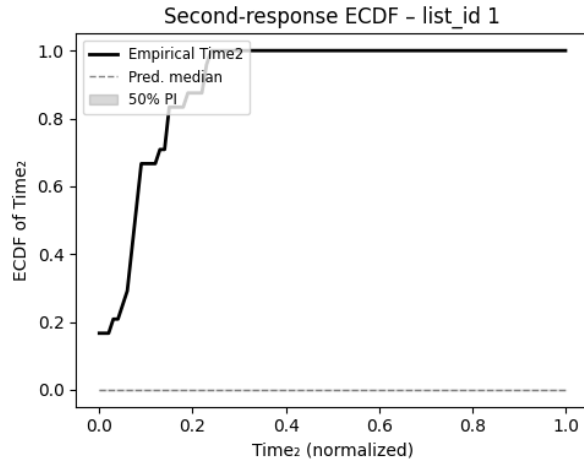


(a) Variation 1

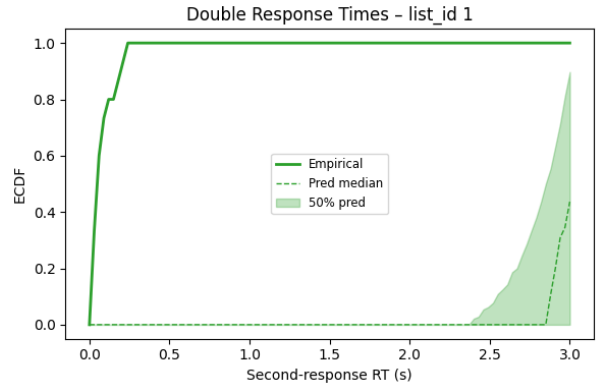


(b) Variation 2

Figure 20: Posterior predictive checks for **A2** second responses. Solid: empirical ECDF of Time2; dashed: posterior–predictive median; shaded: 50% interval.



(a) Variation 1



(b) Variation 2

Figure 21: Posterior predictive checks for **S2** second responses. Solid: empirical ECDF of Time2; dashed: posterior–predictive median; shaded: 50% interval.

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